Depend on vs. *draw on*: Crosslinguistic Insights into Preposition Omission and Semantic Dependencies

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Abstract
While English allows to omit prepositions in elliptical clauses such as *Peter was talking with someone but I don't know* (*with*) *who*, German does not (Merchant 2001, Lemke 2021, among others). Nykiel & Hawkins (2020) argue that the possibility of preposition omission is linked to the relation between the verb and the preposition. If the interpretation of the preposition depends on the verb, omitting the preposition is favored over retaining it in the elliptical clause. This paper tests this prediction for English and German in two acceptability judgment experiments. While V-P semantic dependencies significantly influence the acceptability of preposition omission in both languages, a mirror effect is observed: in English, the predictability of the preposition penalizes its retention, whereas in German, omission becomes less acceptable if the preposition is not predictable.

1. Introduction

In elliptical clauses, English allows to omit prepositions (henceforth, P-omission), as in (1). However, in the German equivalent, P-omission is disallowed, as in (2).

(1) English
Peter was talking with someone, but I don't know (with) who.
(Merchant 2001: 92)

(2) German
Anna hat mit jemandem gesprochen, aber ich weiß nicht, *(mit) wem.
(Merchant 2001: 94)

The sententialist approach and the non-sententialist approach to clausal ellipsis account for this crosslinguistic variation using different lines of reasoning, yet they converge on similar empirical conclusions. In the sententialist literature, Merchant (2001, 2004) argues that P-omission is allowed only if the underlying clause is grammatical in the respective language. For example, English exhibits a grammatical clause, as exemplified in (3), where struckthrough text represents ellipsis. In contrast, the underlying clause for P-omission is ungrammatical in German, as shown in (4).

(3) English
Peter was talking with someone, but I don't know who$_1$ [~~Peter was talking with t_1~~].

(4) German
Peter hat mit jemandem gesprochen, aber ich weiß nicht, wem$_1$ [~~Peter mit t_1 gesprochen hat~~]

However, in the non-sententialist approach, the idea that elliptical clauses require an underlying, syntactic clause is rejected (Ginzburg & Sag 2000, Culicover & Jackendoff 2005). Instead, Nykiel & Hawkins (2020) propose that the possibility of P-omission is linked to processing constraints. That is, retaining the P (henceforth, P-retention) is dispreferred if the interpretation of P is assigned by a V at some distance from PP. This is shown in (5), where the P is assigned by *depend* in the antecedent clause.

(5) English
  A: And you're depending on what?
  B: ($^{?}$On) people going into the booth and changing their minds?
(Nykiel & Hawkins 2020: 424, judgment added)

This paper experimentally tests whether V-P semantic dependencies influence the acceptability of P-omission in German and English. The paper is structured as follows. Chapter 2 introduces the theoretical background that motivates the hypotheses and methodology of the experiments, while reviewing prior experimental work on P-omission. Chapter 3 presents the acceptability judgment experiment conducted on English which is known to allow P-omission. Similarly, chapter 4 presents the experiment on German which is known to disprefer P-omission. Chapter 5 discusses the results and consequences. Finally, chapter 6 summarizes the main findings and provides a conclusion.

2. Literature review

2.1 Deriving fragment answers

Fragment answers, such as speaker B's answer in (6), are "subsentential XPs with the same propositional content and assertoric force as utterances of fully sentential syntactic structures" (van Craenenbroeck & Merchant 2014: 719). In the remainder of the paper, speaker A's utterance is referred to as *antecedent*. The PP in the antecedent clause is referred to as *correlate*. The fragment can either be a PP in the case of P-retention or a DP in the case of P-omission.

(6) English
  A: Peter is talking with someone.
  B: Yes, with his mother. *P-retention*
  B′: Yes, his mother. *P-omission*

In the literature, two main approaches have been proposed to account for the acceptability of fragment answers: the sententialist approach and the non-sententialist approach. In this chapter, I outline the core assumptions of both frameworks, focusing in particular on how they predict the distribution of preposition omission in fragment answers. Based on this discussion, I derive two testable constraints: P-NP COHERENCE, which captures when prepositions must be retained in fragments, and V-P COHERENCE, which accounts for how semantic dependencies between verbs and prepositions impact omission. These constraints, though grounded in different theoretical traditions, make clear empirical predictions that will be tested in the cross-linguistic experiments presented in the next chapter.

2.2 The sententialist approach

The sententialist approach assumes that fragments are the remnants of clausal ellipsis (Merchant 2001, 2004, Griffiths 2019, among others). That is, ellipsis is assumed to involve non-pronunciation of a TP to the exclusion of one phrase, i.e., the fragment. The most well-known approach is the move-and-delete (henceforth, M&D) approach by Merchant (2001, 2004), which assumes that fragments A'-move in order to escape ellipsis. That is, a fragment answer is generated by A’-moving a subclausal phrase to a clause-peripheral position that dominates a clausal node that undergoes ellipsis. The elliptical clause is thus structurally full but phonologically reduced, with the remnant surviving deletion due to its movement to the left periphery.

One of the central arguments for this approach comes from the crosslinguistic distribution of P-omission in elliptical clauses. In his Preposition Stranding Generalization (henceforth, PSG), he proposes that a language allows P-omission in fragment answers only if it allows Ps to be stranded by A′-movement in nonelliptical clauses. Such P-stranding is allowed in English, as shown in (7), but disallowed in German, as shown in (8). Thus, the validity of the PSG seems to be confirmed by English and German, as English is known to allow P-omission, whereas German does not.

(7) English
Who$_1$ was he talking with t_1?
(Merchant 2001, 92, traces added)

(8) German

Wem$_1$	hat	sie	mit	t_1	gesprochen?
who	has	she	with		spoken

(Merchant 2001, 94, traces added)

In sum, the sententialist approach treats fragments as remnants of ellipsis. Fragment answers are only deemed grammatical if the syntactic movement required to derive them is allowed in the respective language. Therefore, the PSG explains crosslinguistic variation in P-omission by appealing to whether a language allows P-stranding under A′-movement.

2.3 The non-sententialist approach

In contrast to the sententialist approach, the non-sententialist approach assumes that there is no underlying structure and that the fragment receives its interpretation from the discourse (Ginzburg & Sag 2000, Culicover & Jackendoff 2005). Within this framework, Nykiel & Hawkins (2020) argue that processing constraints, not syntax, explain the distribution of P-omission. Specifically, they propose that case marking plays a key role: in languages like German, where case is overt, a bare DP without its case-assigning preposition is difficult to process or recover. English, by contrast, lacks productive morphological case in nominal arguments, reducing such processing burdens. That is, Nykiel & Hawkins (2020) propose that P-omission is only acceptable in English, as it has lost its overt case marking system. In contrast, in German exhibits overt case marking in DPs. According to Nykiel & Hawkins (2020), the case-marked DP in the fragment answer can only be properly processed if the case-assigning P is present in the fragment answer. Therefore, only P-retention is acceptable in German, whereas P-omission leads to processing burdens, lowering its acceptability.

Another processing constraint that influences the acceptability of P-omission according to Nykiel & Hawkins (2020) is Minimize Domains (henceforth, MiD), given in (9).

(9) **Minimize Domains (MiD)**
The human processor prefers to minimize the connected sequences of linguistic forms and their conventionally associated syntactic and semantic properties in which relations of combinations and/or dependency are processed. The degree of this preference is proportional to the number of relations whose domains can be minimized in competing sequences or structures, and to the extent of the minimization difference in each domain.
(Hawkins 2004: 31)

According to this principle, linguistic structures are processed more easily when the domains in which dependency relations are computed are minimized. In the case of fragment answers, this principle leads to a prediction about V-P COHERENCE: when a P semantically depends on a V outside its local PP, its interpretation requires "a long-distance processing domain" (Nykiel & Hawkins 2020: 424). This incurs a processing cost, as the parser must retrieve information from the verb to assign a semantic role to the preposition. As a result, such V-P dependencies lead to a strong preference for P-omission.

Hawkins (2000) introduces two tests to determine V-P dependencies: the verb entailment test, given in (10), and the pro-verb entailment test, given in (11), where *i* refers to independent V-P combinations and *d* refers to dependent V-P combinations.

(10) **Verb entailment test**
If [X V PP] entails [X V], then assign V_i. If not, assign V_d.
(adapted from Hawkins 2000: 242)

(11) **Pro-verb entailment test**
If [X V PP] entails [X Pro-V PP] or [*something* Pro-V PP] for any pro-verb sentence listed below, then assign P_i. If not, assign P_d.
(adapted from Hawkins 2000: 242f)

Therefore, three levels of V-P semantic dependencies can be identified: (i) neither test indicates a semantic dependency, (ii) only one test does, resulting in a one-way semantic dependency, or (iii) both tests do, indicating a two-way semantic dependency.

In sum, the non-sententialist approach attributes fragment interpretation to discourse mechanisms and processing constraints. Variation in P-omission is linked to language-specific morphological and cognitive factors, such as the relation between P and the antecedent V.

2.4 Deriving the P-NP COHERENCE

Both the sententialist and non-sententialist approaches predict the observed contrast between German and English with respect to preposition omission in fragment answers. However, they do so based on different, language-specific mechanisms: the sententialist account appeals to syntactic movement, while the non-sententialist account relies on processing-related constraints. What is lacking, however, is a cross-theoretical constraint that can uniformly encode this generalization, regardless of whether one adopts a sentential or non-sentential analysis. This is the role of the P-NP COHERENCE, as formulated in (12). This constraint reflects the idea that a P and its complement form a semantic and syntactic unit, which must be preserved in fragment answers when the correlate is a PP. I adopt the empirical claim shared by both approaches, namely that this constraint holds in German, but not in English.

(12) **P-NP COHERENCE**
The P must be retained in a fragment answer if the correlate is a PP.

The patterns observed in both theoretical accounts point toward a shared underlying constraint on P-retention in German. This motivates the formulation of P-NP COHERENCE, which captures the role of syntactic cohesion in fragment interpretation and sets the stage for examining additional factors that influence P omission.

2.5 Deriving the V-P COHERENCE

While the P-NP COHERENCE captures the syntactic bond between a P and its nominal complement, a second factor identified by Nykiel & Hawkins (2020) concerns the semantic dependency between a V and a P. According to their account, when the interpretation of a P relies on the antecedent V, the resulting structure requires a longer processing domain. This incurs greater cognitive cost, as the parser must access information structure outside the PP to assign a semantic role to the P.

To account for this, Nykiel & Hawkins (2020) propose that such V-P dependencies increase pressure to omit the P entirely in fragment answers. This interacting is captured by the V-P COHERENCE, as formulated in (13), which is predicted to hold cross-linguistically according to Nykiel /&Hawkins (2020).

(13) **V-P COHERENCE** (to be revised)
The presence of V-P dependencies enhances the acceptability of P-omission.

The formulation of V-P COHERENCE adopted here is provisional. I will return to this constraint and revise it in light of the cross-linguistic experimental results.

By shifting focus from syntactic structure to semantic integration across constituents, V-P COHERENCE complements the P-NP COHERENCE and provides a more nuanced account of P-omission. The next section turns to empirical findings that evaluate both constraints and highlights unresolved questions motivating the experimental study.

2.6 Empirical motivation

The effect of semantic dependencies on P-omission has been tested further in following corpus studies. However, interestingly, conflicting trends occur.

In the original corpus study on English, Nykiel & Hawkins (2020) find that the frequency of P-omission and P-retention is at chance if there is no V-P semantic dependency. For one-way semantic dependencies, the odds of P-omission increase 4.4 times, whereas for two-way semantic dependencies, the odds of P-omission increase 4.9 times.

In contrast, in a follow-up corpus study on English, P-retention is found to be more frequent than P-omission for semantically independent Vs and Ps (Nykiel 2025), contrary to the chance-level reported in the original study by Nykiel & Hawkins (2020). Moreover, they find a trend toward increased frequency of P-retention increases if a one-way semantic dependency is present. Though this trend is not statistically significant, it contrasts with the trend found by the original study. Lastly, for two-way semantic dependencies, P-omission is far more frequent than P-retention if two-way semantic dependencies are present. For this level, Nykiel (2025) finds that the odds of P-omission are over 6 times higher.

Hassen & Abeillé (2025) also test the effect of semantic dependencies on P-omission in French. Although they also use a corpus and annotate it in the same way as the original study, no effect of semantic dependencies on P-omission is found (Hassen & Abeillé 2025). In fact, the frequency of P-omission tends to decrease if a semantic dependency is present.

However, there might be several reasons why the effect found in Nykiel & Hawkins (2020) cannot be replicated in the follow-up studies. First, Nykiel (2025) investigates pseudosluicing, which poses a confound when testing the effect of semantic dependencies on P-omission, as P-omission is grammatical under clausal ellipsis but not pseudosluicing (van Riemsdijk 1978). Second, Hassen & Abeillé (2025) limit themselves to *wh*-fragments, which are known to show a higher frequency of P-omission than non-*wh* fragments, e.g., since they allow clefts constructions (Szczegielniak 2008, van Craenenbroeck 2004). Thus, to see a clearer effect of whether P-omission becomes more frequent if semantic dependencies are present, non-*wh* fragments should be tested. Last, corpus studies might not be as reliable because they cannot provide reliable negative evidence.

Therefore, two acceptability judgment tasks are conducted to test the effect of semantic dependencies on P-omission cross-linguistically. The first hypothesis posits that P-omission in fragment answers is significantly degraded in German, but not English, as predicted by the PSG and the P-NP COHERENCE. The acceptability of P-omission in English is tested in an acceptability judgment task by Lemke (2021), who finds that P-omission is slightly more preferred than P-retention, albeit not significantly. The degrading effect of P-omission in German is observed in Molimpakis (2019), Lemke (2021), and Schiele (in press). The present study aims to replicate these findings.

The second hypothesis posits that the presence of semantic dependencies enhances the acceptability of P-omission, as predicted by the V-P COHERENCE. This means that P-omission is more acceptable than P-retention if P depends on V.

In sum, the existing literature offers compelling theoretical and corpus-based insights into the acceptability of preposition omission. Yet differences in methodology and interpretation leave open whether structural and semantic constraints independently shape fragment well-formedness, and how these effects play out cross-linguistically. The following chapter presents two acceptability judgment experiments designed to test these questions in English and German under controlled conditions.

3. Experiment on English

3.1 Participants

32 participants were recruited via *Prolific*, of which 31 participants entered the analysis (for details, see section 3.4). To ensure that only native British English monolingual speakers participated in the study, the pool of participants was filtered on *Prolific* and participants were asked about their native languages. Participants' age and gender were not controlled. Participants were paid 9 GBP per hour for their participation.

3.2 Study design

The experiment was conducted using a 2 (PREPOSITION: retention or omission) x 2 (DEPENDENCY: dependent or independent) factorial within-subject design. The experiment was designed on the research platform *PsychoPy*. Participants were not informed about the variables manipulated. The experiment was conducted online and unsupervised.

The factor PREPOSITION refers to the manipulation in the fragment answer, i.e., whether the fragment answer includes a P or not. That is, one half of the fragment answers within each lexical set were PPs, whereas the other half were DPs. The factor DEPENDENCY refers to the relation between the V and the PP. To distinguish between dependent and independent V-P combinations, Hawkins' (2000) tests were used. If the V-P combination shows no semantic dependency, it is labelled as 'independent'. If the V-P combination shows a one-way or two-way dependency, it is labelled as 'dependent'. In the stimuli used, this distinction is equivalent to whether the PP is an argument or an adjunct (Nykiel 2025). The type of P was the same across each lexical set.

Participants only saw one item, i.e., one dialogue, at a time and were asked to rate the naturalness of Speaker B's answer on a 7-point Likert-type scale, with 1 indicating 'fully unnatural' and 7 indicating 'fully natural'. Each condition was repeated three times, leading to 96 data points per condition. Items were selected from 12 lexical sets, organized into a 4-list Latin square, with a 2:1 ratio of fillers to test items. Additionally, three attention checks were pseudo-randomly interspersed into the trials.

In the study instructions, participants were asked to base their judgments on their intuitions as native speakers and on everyday spoken language, explicitly excluding written language standards. Due to the set-up of the study in *PsychoPy*, participants were required to provide a judgement before continuing the questionnaire. There was no time limit. The experiment took approximately 12 minutes to complete.[1]

3.3 Materials

For each lexical set, four dialogues were constructed. Speaker A's utterance only differed in the V and the complement. Speaker B's answer only differed in whether it included the P or not. The dependent condition of an example lexical set is shown in (12), while the independent condition is shown in (13). The complete list of critical items is included in Appendix 1.

(14) Stimuli with V-P dependency
A: I heard that David depends on someone for money.
B: Yes, on his mother. *P-retention*, *dependent*
B′: Yes, his mother. *P-omission*, *dependent*

(15) Stimuli without V-P dependency
A: I heard that David drew a tattoo on someone for fun.
B: Yes, on his mother. *P-retention*, *independent*
B′: Yes, his mother. *P-omission*, *independent*

To avoid a finality effect as a confounding factor, the correlates in all stimuli were not in sentence-final position. Furthermore, speaker B's answers bore non-contrastive focus, i.e., were introduced by *Yes*. Participants were randomly and equitably assigned to one of four lists. The Latin square design ensured that each participant saw only one item per lexical set.

In addition, each list included 16 standardized filler items, taken from Gerbrich et al. (2017). In addition, 8 additional fillers that were either fully acceptable or fully unacceptable were included to ensure a 2:1 ratio of filler to test items.

Each participant saw three repetitions of the same conditions, totaling 12 critical items per participant, plus 24 filler items, and 3 attention checks. The order of all items was pseudo-random.

3.4 Analysis

As preregistered, participants were excluded if they failed the attention checks. In the present study, no participants were excluded based on the attention checks.

Since the study used standardized fillers whose acceptability ratings were already tested in previous studies (Gerbrich et al. 2017), the expected ratings of the standardized fillers were used to identify outliers based on the sum of square calculation of the difference between expected and observed ratings. Data points that were two standard deviations away from the mean were excluded as outliers (Sprouse et al. 2021). That is, 1 participant was removed as outlier based on their responses to the standardized fillers. All 31 remaining participants entered the analysis, as all of them answered at least two of the attention checks correctly.

[1] The experimental set-up, data analysis plan, and hypotheses were preregistered at https://osf.io/9b4cm

A series of linear mixed effect regression models (LMMs) were performed on the z-scored ratings using *R* (version 4.2.3, R Core Team 2022) and the *lmerTest* package (Kuznetsova et al. 2017). All LMMs displayed a maximal fixed effect structure and the best fitting random effect structure, as determined manually through maximum likelihood testing (*lmerTest*'s anova function). All LMMs have anova-coded predictor variables. The model estimates, t-values, and p-values are generated using *lmerTest*'s summary function. Random intercepts and slopes were included for lexical sets and participants if they improved the model fit (assessed by Likelihood Ratio Tests). The final model predicts the z-scored ratings based on the fixed effect of PREPOSITION, DEPENDENCY, and their interaction. It includes random intercepts for subjects and lexical sets.

3.5 Results

The final data set that entered the analysis included 31 participants. Their responses to the critical stimuli are shown in Figure 1.

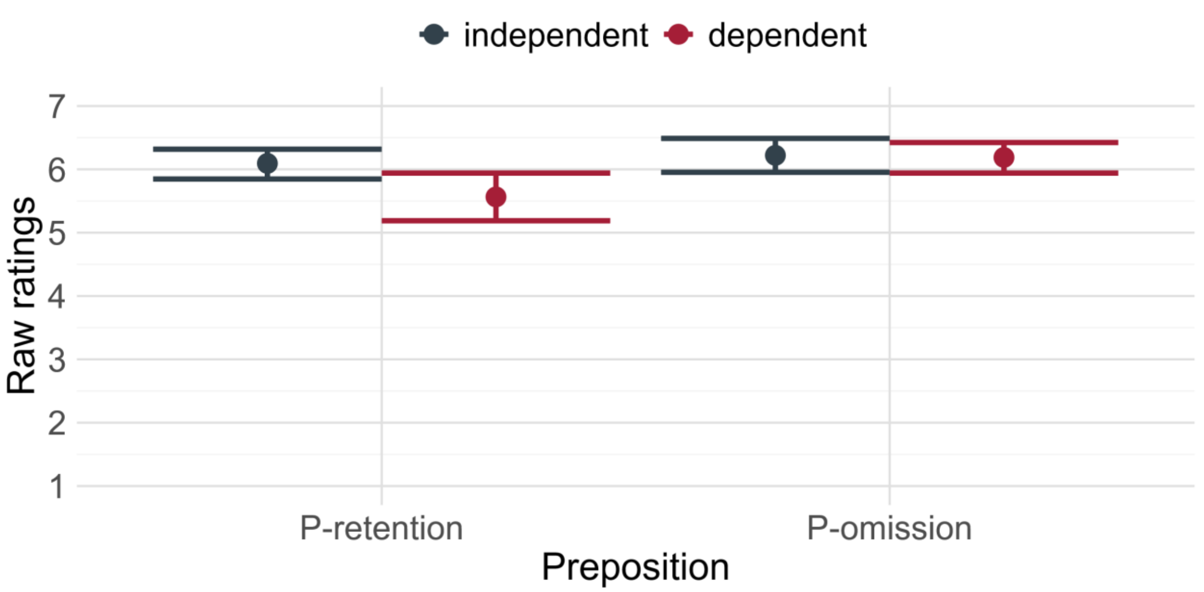

Figure 1

The effect of PREPOSITION is not significant (σ = 0.08, t = 0.65, $p = 0.52$). In contrast, the effect of DEPENDENCY is significant (σ = 0.08, t = -2.90, $p < 0.01$). Crucially, the interaction between PREPOSITION and DEPENDENCY is also significant (σ = 0.12, t = 2.04, $p = 0.04$). That is, the presence of semantic dependencies significantly influences the acceptability of P-omission.

To allow a closer inspection of Nykiel & Hawkins' (2020) predictions, the V-P combinations used in the study are grouped based on Hawkins' (2000) tests, distinguishing between all three levels of V-P dependencies. Figure 2 shows participants' raw ratings across all conditions, when using Nykiel & Hawkins' (2020) coding of semantic dependencies.

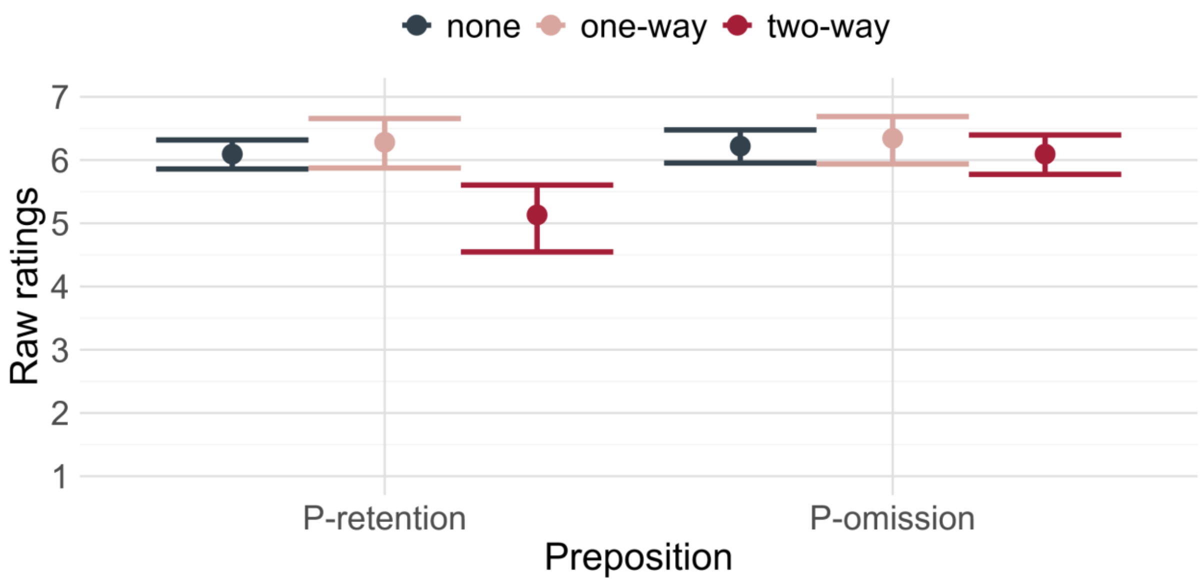

Figure 2

As becomes apparent in Figure 2, the effect of semantic dependencies on P-omission can be seen more clearly if it is two-way. Interestingly, there seems to be no interaction between the V-P combinations without semantic dependencies and with only a one-way semantic dependency. This is discussed in more detail in the following section.

3.6 Interim discussion

The first hypothesis stated that both P-retention and P-omission are acceptable in English. This hypothesis is borne out, as no significant difference is observed for the factor PREPOSITION. Therefore, the prediction that the P-NP COHERENCE does not hold in English is borne out. This finding aligns with Merchant's (2001) PSG and the findings for English by Lemke (2021), who observed a slight preference for P-omission, but found both P-retention and P-omission to be acceptable.

The second hypothesis stated that the presence of semantic dependencies enhances the acceptability of P-omission, as predicted by Nykiel & Hawkins (2020). Although a significant interaction between PREPOSITION and DEPENDENCY is found, this hypothesis is not entirely borne out. Nykiel & Hawkins (2020) predict that P-omission is more acceptable for dependent Ps than independent Ps. However, P-omission receives equal ratings for both dependent and independent Ps. The data show that the presence of semantic dependencies penalizes P-retention, i.e., retaining dependent Ps is less acceptable than retaining independent Ps.

This finding suggests that V-P COHERENCE, as originally formulated, does not capture the English data: semantic dependencies do not license omission, but rather disfavor retention. A revised version of the constraint is thus required.

Furthermore, Nykiel & Hawkins (2020) find an effect of one-way semantic dependencies. However, this finding could not be replicated in the present experiment. The results of the present experiment suggest that only two-way semantic dependencies penalize P-omission.

4. Experiment on German

4.1 Participants

The recruitment and payment were identical to the experiment above, except that only German monolinguals were recruited. Out of the 32 recruited participants, the data of 29 participants entered the analysis (see 4.4 for details). 22 participants were male, while 7 were female. Their age varied between 20 and 70, with a mean of 35.0.

4.2 Study design

The set-up of the experiment was identical to the experiment above. However, the present experiment was designed and conducted via the research platform *Gorilla*.[2]

4.3 Materials

The factors, fillers and the ratio of filler and test items were structurally identical to the experiment above, except that language-specific V-P combinations were utilized to ensure a clear distinction between dependent and independent Ps. The heavy condition of an example lexical set is shown in (14), while the light condition is shown in (15). Moreover, while the English experiment controlled the type of P across each lexical set, the German experiment controlled the type of antecedent V across each lexical set. The complete list of critical items in the German experiment is included in Appendix 2.

(16) Stimuli with V-P dependency
  A: Die Sportler haben sich mit ihrem Trainer über etwas gefreut.
  B: Ja, über den Turniersieg. *P-retention*, *dependent*
  B′: Ja, den Turniersieg. *P-omission*, *dependent*
(17) Stimuli without V-P dependency
  A: Die Sportler haben sich mit jemandem über den Turniersieg gefreut.
  B: Ja, mit ihrem Trainer. *P-retention*, *independent*
  B′: Ja, ihrem Trainer. *P-omission*, *independent*

To avoid case as a confounding factor, accusative and dative case were counterbalanced across the dependent and independent conditions. As in the experiment above, 12 critical items from 12 lexical sets, distributed into a 4-list Latin square design were used. Each participant saw only one item per lexical set. In addition, participants rated 24 filler items consisting of 16 standardized fillers and 8 additional fillers. In addition, 3 attention checks were included.

4.4 Analysis

The removal of participants followed the same criteria as in the experiment above. 3 participants were excluded based on the attention checks. 2 participants were removed as outliers based on their responses to the standardized fillers. In addition, 1 participant who self-identified as bilingual was removed.

The analysis was identical to the experiment above. The model with the best fit as determined by the step-wise model comparison included the same random effect structure as the model in the experiment above. That is, the z-scored ratings are predicted based on the fixed effect of PREPOSITION, DEPENDENCY, and their interaction. It includes random intercepts for subjects and lexical sets.

4.5 Results

The final data set that entered the analysis included 27 participants. Their responses to the critical stimuli are shown in Figure 3.

[2] The experimental set-up, data analysis plan, and hypotheses were preregistered at https://osf.io/h6d5v

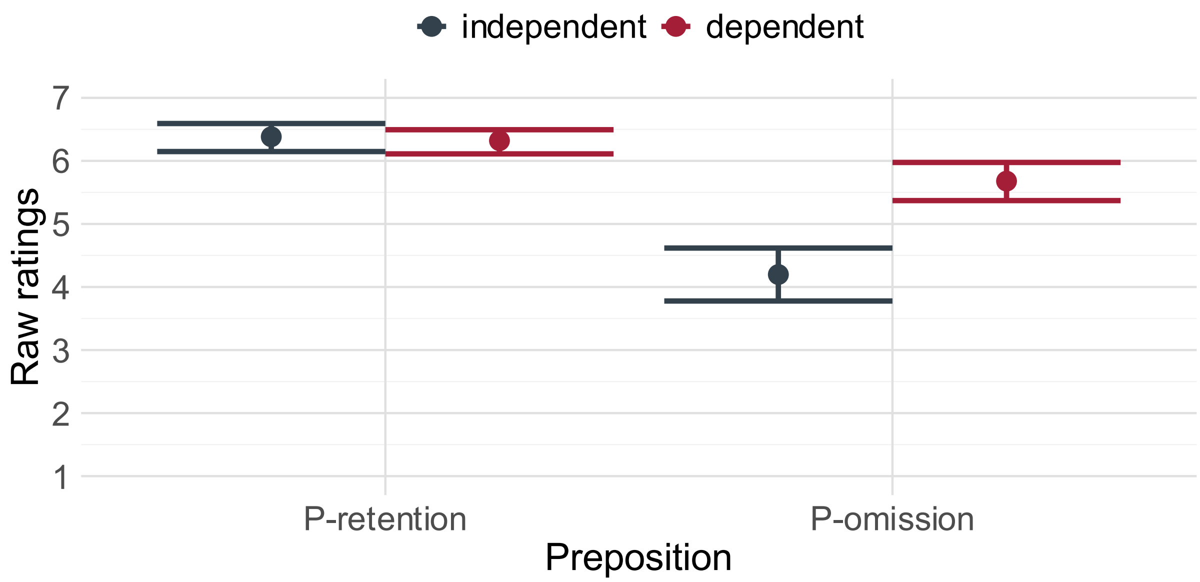

Figure 3

The effect of PREPOSITION is significant (σ = 0.22, t = -4.67, p < 0.01). In contrast, the effect of DEPENDENCY is not significant (σ = 0.22, t = -0.21, p = 0.84). Crucially, the interaction between PREPOSITION and DEPENDENCY is also significant (σ = 0.31, t = 2.48, p = 0.04). That is, the presence of semantic dependencies significantly influences the acceptability of P-omission.

To allow a closer inspection of Nykiel & Hawkins' (2020) predictions, the V-P combinations used in the study are grouped based on Hawkins' (2000) tests. Figure 4 shows participants' raw ratings across all conditions when using Nykiel & Hawkins' (2020) coding.

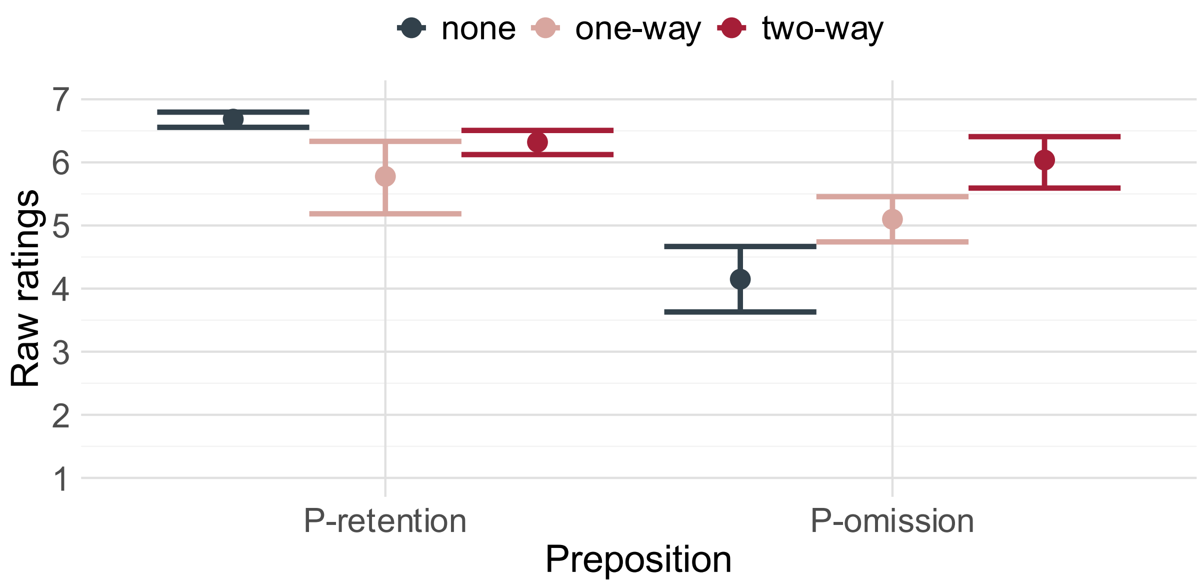

Figure 4

As becomes apparent in Figure 4, the levels of semantic dependencies seem to have an effect. That is, the difference between independent V-P combinations and one-way dependent ones is less pronounced than the difference between independent V-P combinations and two-way dependent ones. In contrast, the differences across DEPENDENCY are comparatively low if the P is retained. This is discussed in more detail in the following section.

4.6 Interim discussion

The first hypothesis stated that P-retention is preferred to P-omission in German. This hypothesis is borne out, as a significant difference is observed for the factor PREPOSITION. That is, omitting Ps has a degrading effect on the acceptability of the fragment answer, whereas fragment answers retaining the P receive the highest ratings. Therefore, the P-NP COHERENCE is borne out in German. This result aligns with Merchant's (2001) PSG and the findings for German by Molimpakis (2016), Lemke (2021), and Schiele (in press).

The second hypothesis stated that the presence of semantic dependencies enhances the acceptability of P-omission, as predicted by Nykiel & Hawkins (2020). Although a significant interaction between PREPOSITION and DEPENDENCY is found, this hypothesis is not entirely borne out. Although the presence of semantic dependencies mitigates the degrading effect of P-omission, P-omission still receives lower acceptability ratings than P-retention. That is, retaining P is generally favored in German, even for dependent Ps. Therefore, the V-P COHERENCE needs to be reformulated for German, as the data suggest that the presence of a V-P semantic dependency mitigates the degrading effect of P-omission in German.

However, Nykiel & Hawkins (2020) predictions regarding the level of semantic dependencies are borne out: the stronger the dependencies are, the more pronounced is its effect on the acceptability of P-omission. That is, omitting two-way dependent Ps is more acceptable than omitting one-way dependent ones. The following chapter discusses these results in comparison to English and previous research findings.

5. Discussion

5.1 Cross-linguistic comparison

The results of both experiments are shown in Figure 5. To allow a comparison between experiments, the standardized fillers of both experiments were compared by matching sample sizes across expected rating levels and running Wilcoxon tests on the subsets. The crucial cut-off points, i.e., level 1, 4, and 7 on the scale, showed no significant differences, supporting the validity of directly comparing the two experiments.

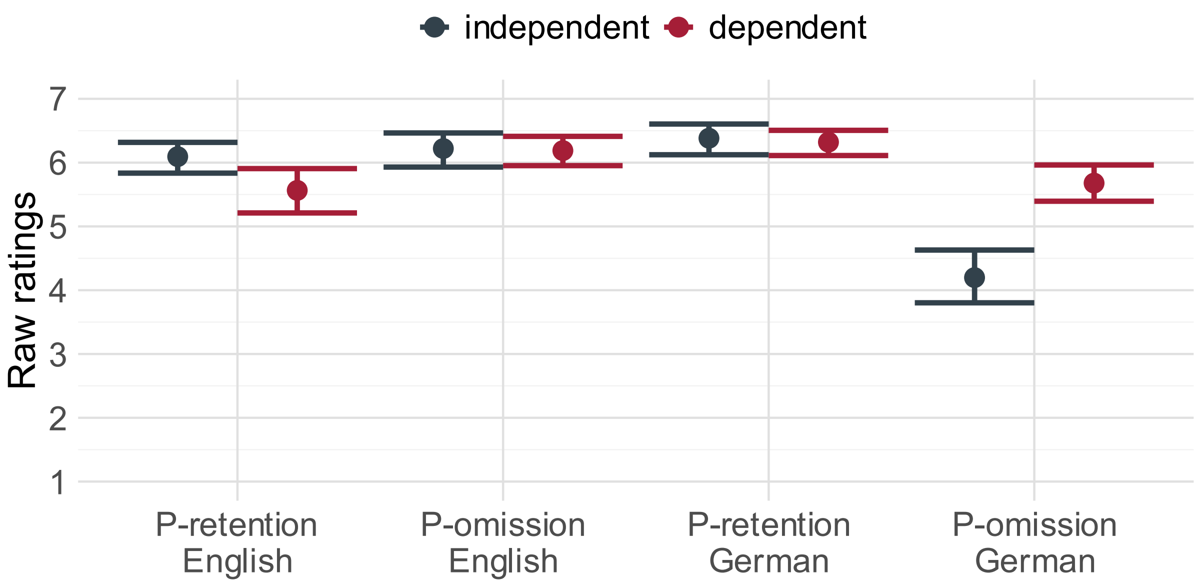

Figure 5

As becomes apparent in Figure 5, the cross-linguistic results for the acceptability of P-omission in fragment answers reveal a mirror pattern. In German, omitting Ps is generally dispreferred, with a stronger penalty for omitting independent Ps than for omitting dependent Ps. In contrast, in English, P-omission is broadly acceptable. In contrast to the German data, retaining a dependent P incurs a penalty in English, while retaining an independent P is acceptable.

5.2 Revisiting the hypotheses

The cross-linguistic results align with expectations regarding their ability to omit Ps. In fragment answers in English, omitting the P is generally acceptable, whereas it leads to degraded acceptability in German. This cross-linguistic variation in the acceptability of P-omission is expected in Merchant's (2001, 2004) sentential approach and Nykiel & Hawkins' (2020) non-sentential approach. Although they use different rationales, they make the same predictions regarding the acceptability of P-omission in English and German.

The degrading effect of P-omission in German also confirms the P-NP COHERENCE. If the constraint is met, no degradation is observed. If it is violated, the acceptability of the fragment answer decreases. As predicted, the P-NP COHERENCE only holds in German, not English.

In addition, the experimental results provide clear evidence for the effect of semantic dependencies on P-omission. However, the observed cross-linguistic pattern concerning V-P semantic dependencies is surprising. According to Nykiel & Hawkins (2020), when a V and a P have a semantic dependency (e.g. *draw on*), omitting the P should be more acceptable, especially in English. This prediction is not borne out in the data. In English, fragment answers receive similar acceptability ratings whether the P is semantically dependent on the V or not. Even more unexpectedly, retaining the P in cases of semantic dependency results in lower ratings, suggesting that its presence is perceived as redundant or dispreferred rather than necessary.

In German, the pattern is even more striking: P-omission is generally unacceptable, even in the presence of V-P semantic dependencies. This suggests that the V-P COHERENCE, as originally formulated, does not capture the cross-linguistic facts, requiring a revised version of this constraint. The following section proposes such a reformulation and explains how it accounts for these results.

5.3 Accounting for the cross-linguistic pattern

The cross-linguistic results suggest that there are two ranked constraints influencing the acceptability of P-omission in fragment answers: the P-NP COHERENCE and the V-P COHERENCE. The first constraint is the P-NP COHERENCE. The P-NP COHERENCE posits that fragment answers must retain the P if the correlate is a PP. This constraint holds only in a subset of languages, e.g., in German, but not English. Depending on the approach, it only holds in non-P-stranding languages or case-marking languages, whereas it does not hold in P-stranding languages or languages that have lost their overt case marking system. Identifying the full typological distribution of languages in which this constraint applies lies beyond the scope of this paper and remains a task for future research.

The data of the present experiments indicate that the V-P COHERENCE has different effects, depending on whether the P-NP COHERENCE holds in the respective language. That is, in languages such as English that generally allow P-omission, retaining P is redundant. This redundancy is particularly notable when P is dependent, as its presence is fully predictable from the V in the antecedent clause, decreasing acceptability.

However, in languages such as German that generally disprefer P-omission, a fragment answer without a P can be processed more easily if P is dependent and thus, predictable. That is, the P and the antecedent V serve as cues, easing resolution. Therefore, the degrading effect of P-omission is mitigated, albeit not equalized. The violation of the P-NP COHERENCE lowers acceptability in German, even if the V-P COHERENCE is not violated. Therefore, omitting dependent Ps is still significantly degraded compared to P-retention. To account for the cross-linguistic variation, the final version of the V-P COHERENCE posits that dependent, i.e., predictable, Ps must be omitted in fragment answers. Note that in German, the P-NP COHERENCE holds, independent of whether the V-P COHERENCE is met or violated.

The V-P COHERENCE is the highest ranked constraint in English, as the P-NP COHERENCE does not apply in English. In contrast, the P-NP COHERENCE is the higher ranked constraint in German, whereas the V-P COHERENCE is the lower ranked constraint. This is illustrated for independent Ps in English in the following Tables.

Candidates with independent P (*draw on*)	**P-NP COHERENCE**	**V-P COHERENCE**	**H**
☞ *on his mother*			0
☞ *his mother*			0

Table 1

Table 1 shows that English fragment answers omitting P (*his mother*) are the optimal candidate, as the P-NP COHERENCE does not hold in English. Similarly, the V-P coherence does not apply here either, as the V-P combination (*draw on*) shown in Table 1 does not include any V-P dependencies. Therefore, P-retention and P-omission are predicted to receive equal acceptability ratings. In contrast, Table 2 shows the candidates with V-P dependencies.

Candidates with dependent P (*depend on*)	**V-P COHERENCE**	**P-NP COHERENCE**	**H**
on his mother	-1		-1
☞ *his mother*			0

Table 2

In Table 2, retaining the dependent P (*on his mother*) violates the V-P COHERENCE. That is, the V-P combination shown in Table 2 (*depend on*) shows V-P dependencies and the P should thus be omitted in the fragment answer. Therefore, the optimal candidate is the fragment answer without the dependent P (*his mother*). The P-NP COHERENCE does not hold in English and can therefore be disregarded here.

In contrast, both the P-NP COHERENCE and the V-P COHERENCE hold in German. In the remainder of this section, I will explain the pattern in the terminology of Serial Harmonic Grammar (Pater 2009, Ryan 2017), extending the framework to syntax (Murphy 2018).

Crucially, violating both constraints reduces the acceptability ratings further. This is illustrated for omitting independent Ps in Table 3. This condition received the lowest acceptability ratings.

Candidates with independent P (*freuen mit*)	P-NP COHERENCE	V-P COHERENCE	H
☞ *mit ihrem Trainer*			0
ihrem Trainer	-2	-1	-3

Table 3

In Table 3, the fragment answer retaining the independent P (*mit ihrem Trainer* 'with their trainer') is the optimal candidate. In contrast, omitting the independent P (*ihrem Trainer* 'their trainer') violates both the V-P COHERENCE and the P-NP COHERENCE. This condition received the lowest ratings across all conditions.

Furthermore, the acceptability is reduced even if only one of the constraints is violated. That is, even if the lower-ranked constraint is met, the acceptability is reduced for violating the higher-ranked constraint. This is seen in the fragment answers omitting dependent Ps in German, as illustrated in Table 4.

Candidates with dependent P (*freuen über*)	P-NP COHERENCE	V-P COHERENCE	H
☞ *über den Turniersieg*		-1	-1
den Turniersieg	-2		-2

Table 4

In Table 4, the fragment answer that retains P (*über den Turniersieg* 'about the tournament victory') is slightly degraded. This outcome reflects a violation of the V-P COHERENCE, despite satisfying the P-NP COHERENCE. Nevertheless, it remains the optimal candidate because the alternative (*den Turniersieg* 'the tournament victory'), which omits the P, breaches the higher-ranked P-NP COHERENCE.

German data, therefore, provide evidence that the two constraints are, in fact. I argue that the P-NP COHERENCE is the higher-ranked constraint in German, whereas the V-P COHERENCE is the lower-ranked constraint. This is evident in the fact that a candidate violating the V-P COHERENCE still receives higher acceptability ratings than one violating the P-NP COHERENCE, indicating that the latter is the higher-ranked constraint in German.

5.4 Summary

The cross-linguistic data reveal that neither the sententialist nor the non-sententialist approach alone can fully account for the observed variation in preposition omission. That is, neither solely syntactic constraints nor solely processing constraints can account for the observed pattern.
That is, neither purely syntactic constraints nor purely processing-based mechanisms are sufficient. Instead, the findings support a hybrid model that integrates both types of constraints, captured in the ranked interaction of the P-NP COHERENCE and the V-P COHERENCE.

As shown throughout this chapter, the P-NP COHERENCE governs whether a P must be retained based on the structural status of its correlate. The mirror pattern observed in the cross-linguistic data suggests that P-NP COHERENCE does not hold in all languages. As becomes clear in this paper, the P-NP COHERENCE, repeated in (18), holds in German, but not in English.

(18) **P-NP COHERENCE**
The P must be retained in a fragment answer if the correlate is a PP.

In contrast, the final formulation of V-P COHERENCE, given in (19), captures the influence of of V-P semantic dependency on the acceptability of P-omission.

(19) **V-P COHERENCE** (final version)
A dependent P must be omitted in a fragment answer if the correlate is a PP.

Together, these constraints explain the mirror pattern across English and German and account for both structure-sensitive and processing-sensitive factors. Importantly, the ranking of these constraints is language-specific: V-P COHERENCE dominates in English, whereas P-NP COHERENCE is ranked higher in German.

This hybrid analysis not only unifies divergent theoretical approaches but also offers a precise and predictive model of fragment interpretation across languages. The conclusion reflects on the implications of this model for broader theories of ellipsis and cross-linguistic variation.

6. Conclusions

Several approaches have sought to explain cross-linguistic variation in the acceptability of preposition omission in fragment answers. Recent studies have suggested that semantic dependencies may play a role in this phenomenon. This paper presented two experiments examining the effect of V-P semantic dependencies on P-omission in German and English. The findings indicate that while such dependencies influence the acceptability of P-omission, neither purely syntactic nor purely processing-based accounts can fully capture the observed cross-linguistic variation. Instead, I have argued for a hybrid analysis that integrates insights from both domains and provided an in-depth analysis of the cross-linguistic variation, following the Serial Harmonic Grammar. Specifically, I proposed two constraints, i.e., the P-NP COHERENCE and the V-P COHERENCE, that jointly predict the conditions under which condition P-omission is permitted. The results from German and English support not only the validity of these constraints but also their relative ranking in the grammar. These findings contribute to our understanding of the interaction between syntax, semantics, and processing in ellipsis phenomena, and suggest that cross-linguistic differences may arise from subtle variations in how languages balance these constraints. Future research might explore whether similar patterns hold in other languages or in other types of ellipsis beyond fragment answers.

Acknowledgments

I thank James Griffiths, Magdaléna Horáčková, and the audiences of ExLing 2024 Paris, the DGfS summer school 2024, and the Big Ellipsis Workshop (as part of ICON 2024) for helpful discussions, feedback, and suggestions. In addition, I thank Ali Can Çiçek and Valentin Pickard for help with generating the stimuli.

Funding information

This research was supported by the German Research Foundation (*Deutsche Forschungsgemeinschaft*), project number 522896622.

Competing interests
The author has no competing interests to declare.

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Appendix 1
The following list includes all critical items used in the experiment on English.

(1) a. A: I heard that David depends on someone for money.
B: Yes, his mother.
b. A: I heard that David depends on someone for money.
B: Yes, on his mother.
c. A: I heard that David drew a tattoo on someone for fun.
B: Yes, his mother.
d. A: I heard that David drew a tattoo on someone for fun.
B: Yes, on his mother.
(2) a. A: I read that the house belongs to someone, despite being vacant for years.
B: Yes, an investor.
b. A: I read that the house belongs to someone, despite being vacant for years.
B: Yes, to an investor.
c. A: I read that the horse was nasty to someone, despite being tame for years.
B: Yes, an inspector.
d. A: I read that the horse was nasty to someone, despite being tame for years.
B: Yes, to an inspector.
(3) a. A: Sue says that Ahmet frequently turns to someone wise for guidance.
B: Yes, his grandmother.
b. A: Sue says that Ahmet frequently turns to someone wise for guidance.
B: Yes, to his grandmother.
c. A: Sue says that Ahmet frequently prays to someone wise for guidance.
B: Yes, his grandmother.
d. A: Sue says that Ahmet frequently prays to someone wise for guidance.
B: Yes, to his grandmother.
(4) a. A: Taylor's friend looked after one of her plants while she was on tour.
B: Yes, a spider plant.
b. A: Taylor's friend looked after one of her plants while she was on tour.
B: Yes, after a spider plant.
c. A: Taylor's friend left after one of her gigs in order to catch a train.
B: Yes, Sunday's gig.
d. A: Taylor's friend left after one of her gigs in order to catch a train.
B: Yes, after Sunday's gig.
(5) a. A: Excessive stress can lead to a mental illness, according to doctors.

B: Yes, a burnout.
b. A: Excessive stress can lead to a mental illness, according to doctors.
B: Yes, to a burnout.
c. A: Excessive stress is eased by listening to calming music, say doctors.
B: Yes, a ballet.
d. A: Excessive stress is eased by listening to calming music, say doctors.
B: Yes, to a ballet.

(6) a. A: Olivia was responsible for someone while her parents were at work.
B: Yes, her little sister.
b. A: Olivia was responsible for someone while her parents were at work.
B: Yes, for her little sister.
c. A: Olivia baked a cake for someone while her parents were at work.
B: Yes, her little sister.
d. A: Olivia baked a cake for someone while her parents were at work.
B: Yes, for her little sister.

(7) a. A: The desolate road led to an old building in the abandoned area of town.
B: Yes, a factory.
b. A: The desolate road led to an old building in the abandoned area of town.
B: Yes, to a factory.
c. A: The heavy rain caused damage to an old building in the centre of town.
B: Yes, a factory.
d. A: The heavy rain caused damage to an old building in the centre of town.
B: Yes, to a factory.

(8) a. A: The journalist referred to someone important to verify the story.
B: Yes, a witness.
b. A: The journalist referred to someone important to verify the story.
B: Yes, to a witness.
c. A: The journalist provided translation services to someone for an interview.
B: Yes, a witness.
d. A: The journalist provided translation services to someone for an interview.
B: Yes, to a witness.

(9) a. A: The couple decided on a new piece of furniture this morning.
B: Yes, the couch.
b. A: The couple decided on a new piece of furniture this morning.
B: Yes, on the couch.
c. A: The cat slept on a piece of newly bought furniture this morning.
B: Yes, the new couch.
d. A: The cat slept on a piece of newly bought furniture this morning.
B: Yes, on the new couch.

(10) a. A: The team made for a pub to celebrate their victory.
B: Yes, the Irish pub.
b. A: The team made for a pub to celebrate their victory.
B: Yes, for the Irish pub.
c. A: The team collected donations for a pub after their victory.
B: Yes, the Irish pub.
d. A: The team collected donations for a pub after their victory.
B: Yes, for the Irish pub.

(11) a. A: Aicha always falls for unsuitable men, unfortunately.

B: Yes, her colleagues.
b. A: Aicha always falls for unsuitable men, unfortunately.
B: Yes, for her colleagues.
c. A: Aicha always brings coffee for someone at work.
B: Yes, her colleagues.
d. A: Aicha always brings coffee for someone at work.
B: Yes, for her colleagues.

(12) a. A: Apparently, the mayor takes a lot of pride in a new building in the town centre.
B: Yes, the shopping centre.
b. A: Apparently, the mayor takes a lot of pride in a new building in the town centre.
B: Yes, in the shopping centre.
c. A: A new pop-up business opened in the centre of town last week.
B: Yes, the shopping centre.
d. A: A new pop-up business opened in the centre of town last week.
B: Yes, in the shopping centre.

Appendix 2

The following list includes all critical items used in the experiment on German.

(1) a. A: Die Sportler haben sich mit ihrem Trainer über etwas gefreut.
B: Ja, den Turniersieg.
b. A: Die Sportler haben sich mit ihrem Trainer über etwas gefreut.
B: Ja, über den Turniersieg.
c. A: Die Sportler haben sich mit jemandem über den Turniersieg gefreut.
B: Ja, ihrem Trainer.
d. A: Die Sportler haben sich mit jemandem über den Turniersieg gefreut.
B: Ja, mit ihrem Trainer.

(2) a. A: Die Ärztin möchte sich in der Tierrettung um Tiere kümmern.
B: Ja, die Schildkröten.
b. A: Die Ärztin möchte sich in der Tierrettung um Tiere kümmern.
B: Ja, um die Schildkröten.
c. A: Die Ärztin möchte sich in einer Einrichtung um die Schildkröten kümmern.
B: Ja, der Tierrettung.
d. A: Die Ärztin möchte sich in einer Einrichtung um die Schildkröten kümmern.
B: Ja, in der Tierrettung.

(3) a. A: Die Studentin hat seit ihrem Umzug oft an jemanden gedacht.
B: Ja, ihre Eltern.
b. A: Die Studentin hat seit ihrem Umzug oft an jemanden gedacht.
B: Ja, an ihre Eltern.
c. A: Die Studentin hat seit einem bestimmten Ereignis oft an ihre Eltern gedacht.
B: Ja, ihrem Umzug.
d. A: Die Studentin hat seit einem bestimmten Ereignis oft an ihre Eltern gedacht.
B: Ja, seit ihrem Umzug.

(4) a. A: Paul hat mit seinem Bruder am Wochenende auf etwas gewartet.
B: Ja, den Zug.
b. A: Paul hat mit seinem Bruder am Wochenende auf etwas gewartet.
B: Ja, auf den Zug.
c. A: Paul hat mit jemandem am Wochenende auf den Zug gewartet.

B: Ja, seinem Bruder.
d. A: Paul hat mit jemandem am Wochenende auf den Zug gewartet.
B: Ja, mit seinem Bruder.

(5) a. A: Er hat sich letzte Woche vor der Chefin über etwas beschwert.
B: Ja, sein Gehalt.
b. A: Er hat sich letzte Woche vor der Chefin über etwas beschwert.
B: Ja, über sein Gehalt.
c. A: Er hat sich letzte Woche vor jemandem über sein Gehalt beschwert.
B: Ja, der Chefin.
d. A: Er hat sich letzte Woche vor jemandem über sein Gehalt beschwert.
B: Ja, vor der Chefin.

(6) a. A: Lea hat gestern Abend bei ihrer Freundin von etwas erzählt.
B: Ja, ihrer Weltreise.
b. A: Lea hat gestern Abend bei ihrer Freundin von etwas erzählt.
B: Ja, von ihrer Weltreise.
c. A: Lea hat gestern Abend bei jemandem von ihrer Weltreise erzählt.
B: Ja, ihrer Freundin.
d. A: Lea hat gestern Abend bei jemandem von ihrer Weltreise erzählt.
B: Ja, bei ihrer Freundin.

(7) a. A: Aicha hat erst seit dieser Woche in einer anderen Straße gewohnt.
B: Ja, der Bahnhofstraße.
b. A: Aicha hat erst seit dieser Woche in einer anderen Straße gewohnt.
B: Ja, in der Bahnhofstraße.
c. A: Aicha hat erst seit Kurzem in der Bahnhofstraße in Stuttgart gewohnt.
B: Ja, dieser Woche.
d. A: Aicha hat erst seit Kurzem in der Bahnhofstraße in Stuttgart gewohnt.
B: Ja, seit dieser Woche.

(8) a. A: Tim hat sich sehr gerne für seine Eltern um ein Haustier gekümmert.
B: Ja, die Katze.
b. A: Tim hat sich sehr gerne für seine Eltern um ein Haustier gekümmert.
B: Ja, um die Katze.
c. A: Tim hat sich sehr gerne für jemanden um die Katze gekümmert.
B: Ja, seine Eltern.
d. A: Tim hat sich sehr gerne für jemanden um die Katze gekümmert.
B: Ja, für seine Eltern.

(9) a. A: Das Kind hat sich in der Bäckerei für eine Süßigkeit entschieden.
B: Ja, einen Donut.
b. A: Das Kind hat sich in der Bäckerei für eine Süßigkeit entschieden.
B: Ja, für einen Donut.
c. A: Das Kind hat sich in einem Laden für einen Donut entschieden.
B: Ja, einer Bäckerei.
d. A: Das Kind hat sich in einem Laden für einen Donut entschieden.
B: Ja, in einer Bäckerei.

(10) a. A: Delfine haben laut den Biologen schon immer zu dieser Klasse gehört.
B: Ja, den Säugetieren.
b. A: Delfine haben laut den Biologen schon immer zu dieser Klasse gehört.
B: Ja, zu den Säugetieren.
c. A: Delfine haben laut den Experten schon immer zu den Säugetieren gehört.

B: Ja, den Biologen.

d. A: Delfine haben laut den Experten schon immer zu den Säugetieren gehört.
B: Ja, laut den Biologen.

(11) a. A: Die Studenten werden sich für ihren Mitbewohner um dessen Haustier kümmern.
B: Ja, dessen Hamster.

b. A: Die Studenten werden sich für ihren Mitbewohner um dessen Haustier kümmern.
B: Ja, um dessen Hamster

c. A: Die Studenten werden sich für jemanden um dessen Hamster kümmern.
B: Ja, ihren Mitbewohner.

d. A: Die Studenten werden sich für jemanden um dessen Hamster kümmern.
B: Ja, für ihren Mitbewohner.

(12) a. A: Wir haben uns dank der Führerschein-App gut auf eine Prüfung vorbereitet.
B: Ja, die Theorieprüfung.

b. A: Wir haben uns dank der Führerschein-App gut auf eine Prüfung vorbereitet.
B: Ja, auf die Theorieprüfung.

c. A: Wir haben uns dank einer App gut auf die Theorieprüfung vorbereitet.
B: Ja, der Führerschein-App.

d. A: Wir haben uns dank einer App gut auf die Theorieprüfung vorbereitet.
B: Ja, dank der Führerschein-App.